Reimagining Decentralized AI

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ABSTRACT

The paper uses aspirations mentioned in the initial research on machine learning decentralization as a lens for examining the current state-of-the-art and exposing opportunities for future innovations. We explore the potential and limitations of decentralized architectures in affording privacy and human agency for end users, competition, and collaboration for wider market and civic players. We then elaborate on the legal and technological developments necessary for decentralized machine learning systems to realize their liberating potential.

CCS CONCEPTS

• **Social and professional topics → Computing / technology policy**; • **Security and privacy → Privacy protections**; • **Computer systems organization → Distributed architectures**; • **Human-centered computing → Collaborative and social computing**; • **Computing methodologies → Machine learning**.

KEYWORDS

machine learning, federated learning, decentralization, governance, privacy, competition

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1 INTRODUCTION

Conventional machine learning (ML) architecture consolidates data, computational resources, and decision-making under the authority of a single entity. Centralization as a mode of production has significant social implications ranging from privacy vulnerabilities [Citron and Solove 2022] to the creation of barriers to innovation and new competitors, as only a handful of institutions and companies can handle this data and computation-intensive mode of production [Luitse and Denkena 2021; Srnicek 2018, 2022]. As the functions of ML in society become more prominent and the social externalities of the centralized mode of production become more evident, decentralizing ML appears to be a promising direction for more inclusive future development. However, ML decentralization, like counterpart concepts such as ML democratization, can be misleading, as they can bear varying interpretations and material expressions [Seger et al. 2023]. Despite ML decentralization's emancipatory appeal, or because of it, some of its incarnations can serve as a fig leaf that conceals and normalizes unjust power relations and harms they claim to solve [Janssen et al. 2020]. This article takes a multidisciplinary approach, combining technological and legal perspectives, to explore ML decentralization's affordances and constraints.

To ground the discussion, the paper focuses on a specific approach to decentralized ML known as Federated Learning (FL). FL operates by decentralizing data and computing and was first presented under this name by Google researchers in 2017 [McMahan et al. 2017]. The computer science literature and corporate marketing materials tend to present FL as a "solution" for two main use cases: The first use case, which is the focus of this paper, involves jointly generating ML models from individuals' private data via computations performed on the users' smartphones or wearable devices [Li et al. 2020b]. Instead of transferring sensitive training data to a central server, only small incremental updates to the main model are transferred [McMahan et al. 2017]. Under this approach, FL has the potential to support the development of applications like next-word prediction, face detection, and voice recognition based on personal data, while not needing to transfer the training data to a centralized server. The other main use case, which is not explored in this paper, includes using FL to jointly train models across multiple organizations or institutions like multiple hospitals or banks, without needing to share their training datasets [Nguyen et al. 2022].

The paper uses aspirations mentioned in the initial research on machine learning decentralization as a lens for examining the current state-of-the-art and exposing opportunities for future innovations. We explore the potential and limitations of decentralized architectures in affording privacy and human agency for end users, competition, and collaboration for wider market and civic players. The affordance approach [Cohen 2019, 2023; Norman 1999] assists in revealing both the emancipatory opportunities that ML decentralization can open and the current obstacles hindering its full realization. Building on these insights, the article identifies the missing technological and legal infrastructures and institutions that could support ML decentralization in better realizing its potential.

Summary of contributions.

- We recognize several key barriers currently holding FL from realizing its potential. We emphasize FL privacy vulnerabilities and the lack of affordance of data contributors to actively participate, benefit and even know about the computing that takes place on their devices, based on their data, their storage space, and electricity. A key takeaway for regulators and practitioners is that decentralized systems cannot be relied upon as the sole means of complying with privacy and data

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protection regulations. Regulators, legislators, and judges must understand that despite the decentralized architecture, the entity directing the computing and benefiting from it should not be exempt from legal and ethical responsibilities concerning model construction. Without such legal policies, decentralization could become a way of avoiding responsibility and evading even the limited obligations that apply today to AI systems.

- The barriers that make it difficult for actors other than giant tech corporations to develop and use decentralized architectures, include the lack of access to decentralized collective data and the lack of preliminary models necessary for distributed computing. We demonstrate that the decentralization of data and computing itself does not challenge the status quo wherein big tech companies have great control over the production and deployment of AI. On the contrary, it may even strengthen it.
- We analyze the potential of a decentralized architecture to address some of the burning contemporary data governance questions. Decentralized ML has the potential to provide built-in protections for individuals' privacy; to enable individuals to have a say concerning the uses of their data to train ML models, and to collaborate with others and produce insights from joint data processing, and in this way to provide radical opportunity for data-based agency and self-rule. Decentralizing ML can also enable a variety of entities to access decentralized personal data for training ML models, in a way that promotes competition and innovation. As data has relational and collective properties, individual control and agency should also be understood as relational. Decentralized architecture can enable individual control and agency implementation by facilitating mediums of conscious data collaborations. This decentralized ML can enable individuals and communities to tell, participate in, and create their life stories through data.
- We elaborate on the infrastructural conditions necessary for decentralized ML systems to realize their liberating potential in terms of privacy and agency and in terms of competitiveness and innovation. The paper discusses two types of supporting technological, legal, and social infrastructures: infrastructures for collaboration in training ML models, and infrastructures for making individual and shared decisions about the use of data. These infrastructures could help FL realize its potential and represent a timely need in data governance in general.

Relation to prior work. Two articles stand at the heart of our analysis. The first is the very influential and highly cited article that coined the name Federated Learning in 2017 [McMahan et al. 2017]. The paper described FL's advantages over centralized ML architectures as reducing privacy risks and tapping the untouched potential of private data held in users' devices [McMahan et al. 2017]. FL has since been the subject of thousands of academic articles, and a number of major tech companies have implemented versions of it [Kairouz et al. 2021; Li et al. 2020a].

In 2015, just two years before the FL paper was published, Reza Shokri and Vitaly Shmatikov sketched a portrait of a decentralized ML system [Shokri and Shmatikov 2015]. Building on previous work on privacy-preserving computing systems, Shokri and Shmatikov's motivation was to enable training models while protecting individuals' raw data. However, from their perspective, the transfer of training from central servers to the data contributors' devices had two additional purposes and advantages besides privacy and security. The first additional objective was to provide individuals, who contribute their data to decentralized processing, with new abilities to shape the training: "in conventional deep learning, data owners have no control over the learning objective and thus no control or even knowledge of what is being inferred from their data…in conventional deep learning, the learned model is not available directly to data owners…Our privacy-preserving deep learning system addresses all of these concerns" [Shokri and Shmatikov 2015]. The second objective was to disrupt Big Tech's monopoly on data and enable a broader range of entities to develop ML models, for their own needs.

Research approach. The paper uses the original, revolutionary ideas behind FL as a lens for examining the current state of the art and for exposing opportunities for future innovations. The multi-angle promise behind decentralized ML—realizing privacy, agency, collaborations, and competition- is a lens to examine where FL stands today compared to the promise—what it delivers, what it does not, how, and to whom. Second, the promise serves as a lens to explore future directions for innovations.

We analyze the relationship between architectures and abstract legal concepts such as privacy agency and competition based on long-standing legal scholarship that sketches the relations between normative considerations and technological architecture and infrastructures [Benkler 2016; Bietti 2022; Britton-Purdy et al. 2019; Cohen 2019, 2023; Fisher and Streinz 2021; Lessig 2007; Reidenberg 1997; Viljoen 2021; Waldman 2022], For examining the decentralized architecture functioning in this matter we use the affordance approach. The concept of affordance focuses on an artifact's possible range of uses[Gibson 1978; Lockton 2005; Norman 1999]. Julie Cohen suggests adopting this perspective in analyzing rights in the digital realm, as "rights also are made available, partly by the constraints and affordances of the physical environment" [Cohen 2019]. Following this perspective, we analyze the infrastructural configurations of FL and ask what they enable or constrain, and for whom. We focus on exercising the right to privacy, and how agency, competition, and cooperation are informed by the FL-built environment's systemic tolerances and prohibitions.

2 FIRST LENS: ORIGINAL PROMISE VS. CURRENT STATE-OF-THE-ART

The confluence of the growing computing power of end-user devices such as cell phones, growing awareness of the importance of data privacy, and the acceleration of privacy regulations, make FL an idea whose time has come. In the very short time since the ideas of decentralized ML systems were first proposed, there has been constant and significant growth in research on FL [Kairouz et al. 2021; Li et al. 2020a, 2021, 2020b]. In parallel, FL has started to make its way into deployment [Bus 2019; Hard et al. 2018; Zhan et al. 2021]. FL is commonly expected to be a game-changer in many domains, such as health or finance, where privacy concerns

and data protection regulations can hinder gaining insights from private information. However, despite vast research efforts and potentially large benefits, so far FL technologies seem to have been implemented nearly exclusively by big tech corporations, who are also the dominant developers of centralized ML. In the following subsections, we examine the promise of FL and compare it with the current state of affairs, mainly based on available research articles.

2.1 The Privacy Reality

A central incentive that motivates and drives the development and adoption of FL is the possibility of producing models from personal data which is problematic to collect directly due to ethical, legal, or organizational constraints [McMahan et al. 2017]. Against this backdrop, FL has been promoted by the industry as a "solution to privacy problems" [McMahan et al. 2017; Pichai 2019] and as a potential mode of compliance in various domains with the emerging regulations [Kairouz et al. 2021]. In this section, we explore whether FL lives up to this promise.

2.1.1 Inadequate Inherent Privacy Protections. The inherent advantage of FL in terms of privacy is that individuals' raw data remains on their devices. Instead, the information extracted from these devices are the results of local training processes—updates to the global model. However, although the original data is not directly transmitted, unless further protections are put in place, those transmitted model updates can (and do!) leak deeply sensitive information about individuals, and sometimes they leak the entirety of the raw data [Bonawitz et al. 2021; Carlini et al. 2018; Melis et al. 2019; Shokri et al. 2017; Wei et al. 2020]. Consequently, while privacy concerns are often given as the main justification for the development and deployment of FL, FL's architecture alone cannot guarantee even minimal protection against leakages of private information [Melis et al. 2019]. To limit the leakage of training data, some deployments of FL use formal noise-addition techniques such as differential privacy [Wei et al. 2020]; these approaches often provide privacy at the cost of reduced model performance or system efficiency [Yu et al. 2020]. Therefore, while decentralization of ML is compatible with privacy improvements, individuals and regulators should remain skeptical of the blind adoption of FL as a single "solution to the privacy problem." A major concern is a potential illusion among individuals, regulators, and perhaps even some technical experts, that decentralization automatically ensures privacy.

2.1.2 Lack of Meaningful Control Mechanism. Another potential benefit of FL architecture is the possibility to create an intervention point for individuals to be consciously involved in decisions around what is computed from their data and how. However, in the contemporary state of affairs, not only is this potential unrealized, but deployment of FL may leave individuals less protected and with fewer opportunities to control the purposes for which and how their data is being used. While papers often refer to the individuals whose devices are part of FL systems as participants, in the current state the computing is not necessarily done for their personal benefit, nor is it conducted with their conscious or deliberate participation. We found no instance of FL allowing substantial control by individuals.

One concern is that despite the lack of inherent privacy protection against leakages [Aono et al. 2017; Melis et al. 2019], FL might be interpreted as providing compliance with the GDPR [Truong et al. 2021] or other similar data protection regulations, in a way that bypasses the need to seek consent from the data subject [Ammad-Ud-Din et al. 2019]. If indeed FL is used as a replacement for consent (flawed as consent and control may be), individuals would not necessarily know that model training is being done on their own devices and own sensitive data, and individuals could have even less awareness and control than they would have had in a centralized architecture.

FL in its current form does not enable meaningful agency for individuals: the individual is passive and cannot influence the model's design in any meaningful way, nor even her contribution to the training process. Participants are passive hosts of processes done on their devices, with their data, using their electricity, without being involved in any meaningful way. It is important to note that in a traditional centralized ML setting, individuals also have no control over the computing and the consequences of using the model, but FL could potentially have opened up new forms of such agency.

2.2 The Competition and Innovation Reality

In 2015, Shokri and Shmatikov suggested that a decentralized architecture could help break the big-tech monopoly over ML. However, in contemporary research and industry discourse around FL, this potential capacity of FL is hardly mentioned; the idea seems to have been forgotten or buried. In current affairs, the leading players who develop and adopt FL are some of the same big tech companies that also dominate centralized ML computing.

While one might have hoped that FL could allow many entities to learn over the same distributed, privacy-protected data, that dream requires gateways to the data on individuals' devices and to a large population of potential participants—–infrastructure that is currently privately controlled. Billions of users' devices, hardware, operating systems, and apps serve as gateways and provide big tech corporations with the infrastructure and standing audience to engage in FL. In contrast, small vendors do not have such infrastructure of their own, nor can they piggyback on the infrastructure of the larger platforms. Ostensibly FL could have been the solution to make that data accessible. However, in the absence of shared infrastructure with an open API, the ability to use personal data for model training remains in the hands of big tech companies. For now, control over the gateway to private data is becoming tantamount to control over private data.

This problem is crucial, as every ML model needs a huge amount of data, but FL needs even more. Since in FL, the data is not directly accessible to the developers, it can be much more difficult to deal with challenges arising from the heterogeneity of the data—for example, finding biases and anomalies. Therefore, a much larger amount of data is often needed under FL to reach the same accuracy levels as traditional machine learning methods [Yu et al. 2020]. Furthermore, if privacy protections such as differential privacy are to be deployed in concert with FL, these can further degrade the accuracy of computations, requiring even more data points for results to remain at the same level of accuracy.

Incumbents hold another significant advantage in building successful FL systems—their preliminary models. Due to the difficulty of creating models without exploring the raw data first (a process that can produce inaccuracies and various fairness problems) and the data-intensive nature of model development, the standard way to deploy FL is to build it on top of an existing, trusted model by updating the trusted model using FL [Yang et al. 2018]. Many big tech companies have existing models to build on; those existing models may have been built using other data sources accessible by the nature of the company's size, or perhaps could even have been built using private data before users' privacy became an ethical or legal barrier. Companies might then continue to build on these existing models and update them with FL. On the other hand, new competitors may lack the preliminary models and data and may have to start from scratch. These new models, if built solely via FL, might suffer from accuracy difficulties resulting from training without access to the raw data, and therefore, new competitors could be at a continuous disadvantage.

Expertise serves as an additional barrier to entry. The development of FL systems requires special expertise, and large companies employ most experts in the field [Srnicek 2022]. Special expertise is required due to several technical challenges: building the technical infrastructure for doing the distributed model updates, managing the heterogeneity of the computing across participants' devices, managing the heterogeneity of the data without seeing it; and, for FL to live up to its promise of privacy, integrating privacy-enhancing technologies. Established companies generally have an advantage in recruiting and training the necessary talent, leading to yet another barrier for new entrants to FL.

Against these barriers, FL currently fails to realize the dream of challenging existing information asymmetries and power relations and opening the information ecosystem to more innovation. In practice, FL may even deepen the data and power gaps in the industry. It could serve as a tool to circumvent data protection and might confer advantages to companies that already have amassed enormous user bases and built substantial infrastructure (via apps, operating systems, or hardware). Given the current reliance in FL deployments on building upon trusted models, strict privacy regulations that restrict future private data gathering and use could actually serve to entrench advantages for companies that have already used data gathered in the past to train trusted models, as new entrants would have even more limited means to develop their base models. In this way, even without direct access to private data, current tech giants might preserve their roles as the controllers of the datafication processes, which gives them "the power to decide which phenomenon gets datafied, by whom, where, and how" [Fisher and Streinz 2021].

3 SECOND LENS – REIMAGINING FL

Architecturally, FL represents a bold departure from the traditional, centralized approach to ML. These architectural changes could open up new horizons for realizing privacy, agency, collaboration and competition in AI development.

3.1 FL's Potential to Mitigate Privacy Vulnerabilities

Decentralized systems can offer several advantages concerning privacy protection. First, they can enable adding built-in safeguards to protect against information leakage and reuse. Second, they can enable individuals to have more direct control, or control through fiduciaries, over the datafication and training process.

As we showed before FL architecture alone does not inherently protect against leakages; however, the combination of the FL architecture together with additional technical privacy guarantees can create a design-based barrier against data leakages.

Just as using individuals' data to train models without their knowledge, understanding, or intentional input can violate their privacy, on the contrary, a system that individuals intentionally contribute to and shape could be supportive of their ability to express and realize themselves. Moreover, FL could offer a radical opportunity for data-based agency and self-rule, provided that an FL-based interface enabled users to be active, informed participants and not summed up as a passive consent-based feature. Facilitating individuals' agency involves creating technological and legal structures that enable individuals and communities to tell, participate in, and create their life stories through data. The IEEE defines agency in the digital arena as an ecosystem where individuals' terms and conditions can be recognized and honored at an algorithmic level [Chatila and Havens 2019]. Additionally, an interface that permits individual interventions can be a point of deliberate friction [Elkin-Koren et al. 2022; Goodman 2020; Lessig 2007; McGeveran 2013]. It can slow down data utilization and promote conversation between data contributors and those managing and profiting from the computing. Even if the interface is performative only, it can serve as a meaningful reminder of the presence of computing on individual devices and data taking place behind the scenes.

However, despite its allure, reliance on individual control and agency should raise a warning flag. Using an FL architecture does not resolve the main shortcomings of the control-based approach to privacy: the lack of time, knowledge, and motivation of individuals to make decisions in a very complex field, and the limitations of individual control as a solution to the structural and relational privacy vulnerabilities of the information economy[Commission 2022; Richards and Hartzog 2018]. Moreover, physical control alone is not sufficient for enabling substantial control, as many processes occur on personal devices without individuals' knowledge or meaningful consent [Janssen et al. 2020]. To mitigate this problem, FL's control mechanism should be supported by legislation or regulation that will offer diverse proxy frameworks for making decisions about data, which would facilitate and simplify individuals' participation, as we describe later. These frameworks should offer among other things, mechanisms of control through representatives or collective control, through, for example, data cooperatives[Gordon-Tapiero et al. 2022, 2023]. Such legal-social-technological structures may reduce the gap between the promise of individual control, and the reality where power relations result in individuals having no control.

3.2 Opening Options for Horizontal Social Collaborations

Centralized ML architecture is based on horizontal analysis at the population level, which is done from a top-down view, centering the needs, interests, and values of the platforms, but not necessarily those of the relevant populations [Viljoen 2021]. If we take the decentralized structure of FL one step further, one can imagine how it might enable novel forms of collaboration and personalization, by establishing options for conscious and deliberate collaboration between individuals to train models.

Consider, for example, next-word prediction systems on mobile phones, one of FL's current flagship use cases [Yang et al. 2018]. This model type is produced from the raw material of individuals' previous typing and choice of words, to help with future typing and expression. However, individuals from different demographics are likely to have different choices of words due to diverse generational, linguistic and cultural nuances. Therefore, a more personalized prediction pattern may be appropriate [Tan et al. 2022]. In a traditional centralized architecture, data from numerous individuals is gathered and analyzed collectively and any model personalization relies on subgroup identification, detected from above by identifying similar patterns in big data. Under this approach, personalization of the model that directs individual expression is done without their knowledge and active involvement [Shadmy 2022]. On the other side of the spectrum, in a local or edge computing architecture, companies create preliminary models (potentially in a centralized manner) but then transfer them to individual users' devices for use, where personalization can take place, using only the individual's data [Yu et al. 2020]. Under this architecture, individuals have strong privacy guarantees but are also isolated, as only one person's data may be insufficient for personalization.

FL appears to offer the best of both worlds[Shokri and Shmatikov 2015]. In FL on the one hand, the training is done locally, but on the other hand, the improved model aggregates the training results of many individuals. It invites imagining new possibilities of personalization, in which individuals have more agency in designing the engine that constitutes part of their means of expression. Philosophical reflections on the nature of expression and language can help to imagine such new personalization possibilities. The philosopher Charles Taylor has argued that language constitutes personality and consciousness [Taylor 2016]. Therefore, since next-word prediction systems complement human expression by producing linguistic outputs that have a constitutional role, there is justification for allowing individuals whose expressions are produced by the system to have some control over its design. However, Taylor does not see language as merely an individual medium. He elaborates on language's fundamental social component, as meaning is produced only by communication between people: "To have a language is not only to have a device to encode information, but also to dwell in a social space where one can converse with others, and where persons hold each other accountable for correctness" [Taylor 2016]. Therefore, individual control of personalization is not enough; the constitutive social aspect of language must also be brought to light.

Accordingly, one can think of collaborative learning mechanisms by which FL might support both individual agency and the constitutive social aspect of language. First, individuals could choose groups with which their private data would be processed, to create a sub-model that reflects their dialect and social circles. For example, immigrant women from the MidMLe East, feminist Los Angeles residents, etc. However, such an approach has its risks: privacy risks and the concomitant technical challenges, polarizing potential, and the question of who would be empowered to define these sub-groups, since the very definition of the groups is already a choice with social and political significance [Boyd and Crawford 2012]. Another approach could be to provide individuals with technical, legal, organizational, and design opportunities to utilize the data they generate in interactions with various platforms, for their purposes. With the help of FL, it is possible to imagine a system that might allow existing organizations and communities—such as feminist, religious, or ethnic organizations—to develop models based on their consenting members' data. A more ambitious structure is market or civic arenas, which might enable the creation of ad hoc data-based initiatives to produce new models from individual personal data. This could enable individuals without existing communities that express their values to find like-minded individuals and build models that reflect their shared worldview. Such a proposal requires building new infrastructures, which will be discussed later. In the meantime, our point here is to recognize the potential and possibilities that decentralized ML opens up for the personalization of models based on cooperation and collaboration instead of a top-down statistical view.

Providing such agency could also help address the epistemic injustice inherited in traditional ML, where data science has more to say about the individual and the community than the subject's testimony [McQuillan 2019]. FL can potentially mitigate this risk of epistemic injustice by allowing individuals to choose the community or the members of the groups within which their data is processed, resulting in models that may better reflect their perspectives.

In principle, these possibilities could also be implemented in a traditional centralized structure of ML (given some changes), but the development of decentralized systems offers an opportunity to rethink the production processes that have already solidified in the traditional architecture and to examine how new affordance of individual and joint agency could be incorporated.

3.3 Potential for Promoting Competition

In their 2015 article, Shokri and Shmatikov suggest that FL could opening society's collective data to a wide range of organizations wishing to train models. Such an open, distributed approach could allow small startups, nonprofits, or governmental organizations to overcome network effects and create ML models without needing massive databases and massive computing power (as computations are performed at least partially on the end-user devices).

In an ecosystem where the quality of the product depends on the scale of the data and computing, a possible channel for enabling competition and innovation is by enabling cooperation, allowing individuals, communities, and small vendors to collaboratively create larger datasets for training models [Siddarth et al. 2021; Zygmuntowski et al. 2021]. Based on Elinor Ostrom's theorization of the value of cooperation and collective management of collectively owned resources [Ostrom 1990] and Yochai Benkler's extension of

these ideas to digital markets [Benkler 2006, 2011], Elettra Bietti suggests adopting cooperation as a central principle for regulating digital markets [Bietti 2022]. FL is an architecture that could enable such cooperation as FL enables multiple actors to build a common, robust ML model without sharing raw data. By allowing data to be used without transferring it to central servers, FL opens up opportunities for small vendors with limited and homogeneous data to collaborate with others and gain insights from data without needing to transfer it. This architecture also potentially allows for data generated through an individual's interaction with a large platform to be reused by the individual for other purposes. As a result, the FL model could help small vendors overcome their inherent disadvantages in the data market. From this point of view, the story of FL could be a story of challenging the entrenched power structures in the field.

4 PATHS FORWARD

As we have seen, decentralizing data and computing does not necessarily challenge this status quo. On the contrary, it may even strengthen it. This finding is unsurprising given the vast body of literature critically examining computerized decentralization [Agre 2003; Benkler 2006; Golumbia 2016; Hintz 2014].(Mathew 2016) A decentralized system can only guarantee its liberating potential in terms of privacy and agency for individuals and in terms of competitiveness and innovation, with the establishment of supporting technological, legal, and social institutions and infrastructures. This part briefly discusses two types of such arrangements: an infrastructure for collaboration in training models, and an infrastructure and institutions for making individual and shared decisions about the use of data. These infrastructures could help FL realize its potential, but they also represent a timely need in data governance in general.

4.1 Decision-Making Mechanisms

Accessible infrastructure and interfaces for individual and group-based decision-making are necessary to ensure that decentralization is not only a technical condition but also reflects the decentralization of power and authority. Enabling individuals to participate in decision-making regarding using their personal data for ML model training could assist in limiting the powers of giant digital platforms. Additionally, it reflects respect, protection, and fulfillment of individual rights: the right to privacy, freedom of expression and association, self-rule, and equal respect [Shadmy 2022]. Individuals' agency in digital environments does not mean only conditioning the collection and processing of private data on informed consent. It also means enabling individuals to have a voice in what is computed using their data, and how and who gets benefits and access to the results.

However, calls for individual participation in decisions that affect them in the digital realm often collide with real-world cognitive, social, and structural barriers, and with the risk that these mechanisms would help those who benefit from the computation to avoid responsibility for its harms [Richards 2021; Solove 2023]. Taking into account these difficulties on the one hand, and the justifications for participation on the other hand, the law could provide pluralized proxy structures and institutions with predefined decision-making mechanisms that would enable and support different modes of data collaboration, according to the type of relationship governed.

In parallel, this goal raises interesting technical challenges in terms of collective decision-making around data.

4.2 Data Gateway Infrastructures

If one wishes to enable pluralized cooperation and collaboration in developing FL models, we will need new technical and legal infrastructures for shared and decentralized training on personal data. To support such an effort, legislation could establish rights and corresponding obligations. This includes portability rights that would enable individuals to use their personal data and enable third parties to use it, even if it was created during individuals' activities on other platforms. Interoperability requirements could further compel platforms to use standards and protocols that would enable the exchange and the mutual compatibility of the data [Wegner 1996]. Future legislation could also consider mandated competitor access to models created based on personal information, thus allowing new competitors access to initial models that could allow new models to be developed and updated using FL, thus enabling competition in the market. European legislative efforts addressing various data-sharing options, including the GDPR, DMA, DGA, and EU Data Act provide initial steps toward granting some of these abilities.

In synergy with legislation that pushes tech companies to make their FL infrastructure more open and interoperable, technical developments could help ensure that individuals and groups are poised to leverage these new rights and abilities. Through a combination of data-sharing, legal and technical structures that could grant broad access to data that is currently concentrated mainly under the control of big platforms, and interoperability and common standards for data storage, FL technology could help to advance both public and community-based initiatives to use data for the common good and the preservation of open markets receptive to small, independent entrepreneurs.

5 CONCLUSION

In his article "Degrees of Freedom, Dimensions of Power," Yochai Benkler demonstrates how technological structures and architectures have enabled the concentration of power on the Internet and the emergence of new gatekeepers [Benkler 2016]. Following this gaze, in this article we point to a unique moment at which ML's technological structures and the power relations they afford and constrain have just begun to solidify. FL invites us to imagine a world in which data is made available more widely, more equitably, and with proper privacy protections. It could enable exciting new collaboration and agency around data, and new possibilities for innovation and competition. However, at the same time, the architecture creates risks of empty promises entrenching surveillance, the passivity of the users, and the unequal distribution of the benefits of computing. Key to determining which path we follow will be a reckoning with some of the key next-generation dilemmas in data governance, demanding both technological and legal innovation.

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